$$A = \begin{bmatrix} | & | & | & | & | \\ p_1 & f_1 & p_2 & p_3 & f_2 \\ | & | & | & | & | \end{bmatrix} \qquad P = \underbrace{\begin{bmatrix} | & | & | \\ p_1 & p_2 & p_3 \\ | & | & | \end{bmatrix}}$$

no col is a lin. comb. of any other col.

$\mathrm{Col}(A)$ = set of all linear comb of cols of A.
$= \{ A\underline{x} : \underline{x} \in \mathbb{R}^5 \}$.

$$\begin{bmatrix} | & | & | & | & | & | \\ p_1 & f_1 & p_2 & p_3 & f_2 & A\underline{x} \\ | & | & | & | & | & | \end{bmatrix} \therefore \qquad \begin{bmatrix} | & | & | & | \\ p_1 & p_2 & p_3 & A\underline{x} \\ | & | & | & | \end{bmatrix}.$$

$A\underline{x} = PC\underline{x} = P\underline{y}$
(where $\underline{y} = C\underline{x}$).

$$A = \begin{bmatrix} | & | & | \\ p_1 & p_2 & p_3 \\ | & | & | \end{bmatrix} \begin{bmatrix} 1 & \text{first 3 elts. of Second col of rref(A),} & 0 & 0 & \text{first 3 elts of last col of rref(A)} \\ 0 & & 1 & 0 & \\ 0 & & 0 & 1 & \end{bmatrix}.$$

$A = PC$ (C = first 3 rows of rref(A)).

All max linear independent sets in a linear space L have the Same size.
Proof by contradiction.
Suppose $V = \{\underline{v}_1 \ldots \underline{v}_n\}$ and $W = \{\underline{w}_1 \ldots \underline{w}_m\}$ are both max linearly idpt. Without loss of generality assume $n > m$.

$$W' = \begin{bmatrix} | & & | & | \\ \underline{w}_1 & \cdots & \underline{w}_m & \underline{v}_1 \\ | & & | & | \end{bmatrix}$$

$$V = W \underbrace{\begin{bmatrix} \alpha_{11} & \cdots & \alpha_{1n} \\ \alpha_{21} & & \alpha_{2n} \\ & & \vdots \\ \alpha_{m1} & & \alpha_{mn} \end{bmatrix}}.$$

Some col must be free, since $n > m$.

$\beta_1 \underline{c}_1 + \beta_2 \underline{c}_2 \cdots + \beta_{k-1} \underline{c}_{k-1} - \underline{c}_k = 0$.

$\underline{c}_k$ = linear comb of $c_1 \ldots c_{k-1}$. $\overset{\text{say}}{=}$ $\underline{\beta_1 \underline{c}_1 + \beta_2 \underline{c}_2 \cdots + \beta_{k-1} \underline{c}_{k-1}}$

$$\therefore \underbrace{\begin{bmatrix} \alpha_{11} & \cdots & \alpha_{1n} \\ \alpha_{21} & & \alpha_{2n} \\ & & \vdots \\ \alpha_{m1} & & \alpha_{mn} \end{bmatrix}}\underbrace{\begin{pmatrix} \beta_1 \\ \beta_2 \\ \vdots \\ \beta_{k-1} \\ -1 \\ 0 \\ \vdots \\ 0 \end{pmatrix}}_{\to\, \underline{z}} = \underline{0}.$$

$$V\underline{z} = W \underbrace{\begin{bmatrix} \alpha_{11} & \cdots & \alpha_{1n} \\ \alpha_{21} & & \alpha_{2n} \\ & & \vdots \\ \alpha_{m1} & & \alpha_{mn} \end{bmatrix}} \underline{z}$$

Null space of A

$$\text{null}(A) = \{ \underline{z} : A\underline{z} = \underline{0} \}.$$

linear space. $\underline{z}_1, \underline{z}_2 \in \text{null}(A)$.

Does $\alpha \underline{z}_1 + \beta \underline{z}_2 \in \text{null}(A)$?

$A(\alpha \underline{z}_1 + \beta \underline{z}_2) = \alpha A\underline{z}_1 + \beta A\underline{z}_2 = \underline{0}$

$$\begin{array}{ccccc} c1 & c2 & c3 & c4 & c5 \end{array}$$
$$\begin{bmatrix} 2 & 0 & 2 & 1 & 3 \\ 1 & 0 & 1 & 0 & 1 \\ -1 & 1 & -2 & 0 & 0 \end{bmatrix} \xrightarrow{\text{rref}} \begin{bmatrix} 1 & 0 & 1 & 0 & 1 \\ 0 & 1 & -1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 1 \end{bmatrix}$$

$c1 - c2$. $c1 + c2 + c4$.

$\to \underline{c3} = \underline{c1} - \underline{c2}$ $\underline{c1} - \underline{c2} - \underline{c3} = \underline{0}$.

$$\therefore \beta(\underline{c1} - \underline{c2} - \underline{c3}) = \begin{bmatrix} 2 & 0 & 2 & 1 & 3 \\ 1 & 0 & 1 & 0 & 1 \\ -1 & 1 & -2 & 0 & 0 \end{bmatrix} \begin{pmatrix} \beta \\ -\beta \\ -\beta \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

$\underline{c5} = \underline{c1} + \underline{c2} + \underline{c4}$. $\underline{c1} + \underline{c2} + \underline{c4} - \underline{c5} = \underline{0}$.

$$\underline{c1} + \underline{c2} + \underline{c4} - \underline{c5} = \underbrace{\begin{bmatrix} 2 & 0 & 2 & 1 & 3 \\ 1 & 0 & 1 & 0 & 1 \\ -1 & 1 & -2 & 0 & 0 \end{bmatrix}}_{A} \begin{pmatrix} 1 \\ 1 \\ 0 \\ 1 \\ -1 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

$A\underline{x} = \underline{b}$

$A\underline{z} = \underline{0}$ $\underline{z} \in \text{null}(A)$

$A\underline{x} + A\underline{z} = \underline{b} + \underline{0} = \underline{b}$

$= A(\underline{x} + \underline{z})$

$S = \{ \text{set of all solns for } A\underline{x} = \underline{b} \}$.

$= \{ \underline{x}_p + \underline{z} : \underline{x}_p \text{ is a vector satisfying } A\underline{x}_p = \underline{b},\ \underline{z} \in \text{null}(A) \}$.

Assume $\underline{b} \neq 0$.

Let S be the set of all solns for $A\underline{x} = b$.

Is S a linear space?